ScienceDirect

Biosystems Engineering
Volume 130, February 2015, Pages 60-71

Review

Challenges in the development of Precision Beekeeping

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Highlights

- Development of data collection, data analysis and application phases is considered.
- Decision Support Systems (DSS) are a mid-term task for Precision Beekeeping (BP).
- New classes of DSS controlled remote electronic devices are suggested.
- Industrial experiments are needed to estimate the economic effects of PB systems.
- Interdisciplinary collaboration is needed for new counterintuitive PB solutions.

Precision Beekeeping (PB), a sub-branch of Precision Agriculture, is an apiary management strategy based on the monitoring of individual bee colonies to minimise resource consumption and maximise the productivity of bees. Although there are sufficient technical means and industrial products for the practical execution of PB, the process is slow due to the differing states of development of three implementation phases: data collection, data analysis and application.

The technical execution, adequacy of earlier and current solutions and development perspectives are analysed to prioritise different challenges for PB. The development of decision support systems, which is usually a stumbling block in Precision Agriculture, is suggested to be a mid-term task. In the long term, specific DSS-controlled electronic devices should be developed to enable new functionalities for PB. Specific classes of actors are proposed to operate within future PB systems for the automatic execution of suggestions made by decision support systems, as well as for diagnostic purposes.

PB systems should be optimised by searching for appropriate combinations of different sensors, and corresponding decision support systems must provide convenient, reliable and cost efficient solutions. The development and specification of PB systems should consider business interests, distance to the apiary, expected risks and other peculiarities.

Introduction

Precision Agriculture (PA) methods have been developed and adapted for a number of agriculture-related sub-branches such as Precision Farming, Precision Livestock Farming, Precision Viticulture, Precision Horticulture, and Precision Forestry (Zacepins et al., 2012, Zacepins and Stalidzans, 2013).

Recently, Precision Beekeeping (PB) or Precision Apiculture has been defined as an apiary management strategy based on the monitoring of individual bee colonies to minimise resource consumption and maximise the productivity of bees (Zacepins et al., 2012). The PB approach aims to reduce the stress and unnecessary activities of bee colonies as well as the waste of resources. This is achieved through the timely and adequate reaction of beekeepers at the individual colony level in cases of necessity. The aim of the proposed solution is not to replace but rather to support the beekeeper, who will always remain the crucial factor in the good management of honey bees.

How could the adaptation of PA methods help in beekeeping? There are different tasks for PB at the bee colony and apiary scales both in passive and active periods of honey bee colony development (Ambrose, 1992, Schmickl and Crailsheim, 2004, Stalidzans and Berzonis, 2013). It would be an industrial advantage to remotely detect at colony level:

- developmental states of bee colonies (Stalidzans & Berzonis, 2013);
- events that may require the beekeeper's action, including swarming/pre-swarming state, extreme nectar flow, queenless state, broodless state, dead colony, starving, and first cleaning flight in spring;
- diseases, including colony collapse disorder (CCD) (Cox-Foster et al., 2007, Van Engelsdorp et al., 2008).

While the first and second groups of diagnostics tasks are mostly interesting for a particular beekeeper, the early diagnostics of disease, particularly infectious disease, can prevent the losses of bee colonies across wider areas. It would be very important to be able to detect and follow the dynamics of CCD (Cox-Foster et al., 2007, Van Engelsdorp et al., 2008) and diseases such as varroosis. Insights into the dynamics of disease spread and other problems, such as those that may be associated with global warming, could be achieved if the daily activities of many colonies are recorded and stored over several years (Mezquida & Martinez, 2009). Early disease diagnostics would also reduce the contamination of bee products with different residues of active substances used for disease treatment.

PB is an emerging research direction that is supported by scientists from different countries. The ICT-AGRI (ICT-AGRI Secretariat, 2013) ERA-net consortium in the European Union has supported a project called "*ITApic: Application of information technologies in Precision Apiculture*" during 2013–2016 (ITAPIC Latvia, 2013). Another project entitled "The Swarmonitor" aims to develop a diagnostic tool for honey bee colonies by monitoring vibrations in the hive (Swarmonitor, 2014). These types of projects will facilitate the implementation of PB in the future, but there are still many challenges for the successful implementation of PB into beekeeping practices.

The aim of this paper is to analyse the current status, challenges and development perspectives of Precision Beekeeping. Similarly to the three phase cycle of Precision Agriculture (Terry, 2006, p. 224), Precision Beekeeping can also be split and analysed in three phases: data collection (Section 2 of this article), data analysis (Section 3) and application (Section 4) (Zacepins et al., 2012). During the data collection phase, measurements from bee colonies and the environment are collected. The data analysis phase draws conclusions regarding bee colony behaviour and activity trends based on measurement data, predefined models and expert knowledge. In the application phase, decisions are made and actions undertaken based on data analysis for improving apiary performance.

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Section snippets

Data collection phase in Precision Beekeeping

The main objective of the data collection phase of PB is the development of real time and online tools for the continuous monitoring of apiaries. These tools operate at least partly at the individual colony level and use solutions based on automatic or automated information technology (Zacepins et al., 2012). Various technologies can be applied to monitoring the bee colony (Meikle & Holst, 2014), but only the most informative measurements are feasible for the PB approach. In this section, the …

Data analysis phase in Precision Beekeeping

Currently, researchers and individuals view monitored data and correlate it with the colony status (state) by inspecting the hive. Only after the meaning of the data becomes well-understood can automated analysis and recognition algorithms be developed for data interpretation, which can then be used to inform the beekeeper of the colony state (Linton, 2012). Recognition rules have to be tested for statistical significance in connection with the resolution and accuracy of the measurements. This …

Application phase in Precision Beekeeping

The third phase is application, and in the case of beekeeping this cannot be significantly automated because beekeeping is mostly performed manually. The literature reports only a few automated solutions that can be applied in practice.

Automated indoor bee wintering (McCutcheon, 1984, Zacepins and Stalidzans, 2012) gives the opportunity to control the microclimate, which influences the behaviour of indoor wintered bees. The temperature can be changed remotely and can be increased if brood …

Conclusions

The currently available PB systems are not widely used, possibly due to a combination of several reasons: 1) the benefit/investment ratio is unclear; 2) the use is complicated; 3) sensors and electronics require additional operational costs; and 4) there may be differences in beekeeping practices due to the subspecies of *A. mellifera*, climate and other peculiarities.

Currently, the most developed and advanced phase of PB is data collection. Nonetheless, this phase will remain underdeveloped …

Acknowledgements

The scientific research and publication are partly supported by the ERA-NET ICT-Agri Project “Application of Information Technologies in Precision Apiculture (ITAPIC).” The local agreement number is Z/13/1128.

The scientific research and publication are partly supported by the SIA “TIBIT”.

The authors would like to thank U. Richter, P. Ahrendt, S. Tekin and M. Durgun for their comments and important suggestions. …

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2022, Online Information Review

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Toward an intelligent and efficient beehive: A survey of precision beekeeping systems and services
2022, Computers and Electronics in Agriculture

> *Citation Excerpt :*
>
> …During winter, there are fewer bees and fewer resources than during summer, honey stock, which is the food consumed by bees, can be estimated thanks to this metric. Abnormal weight variations can also be correlated to incoming swarming events (Zacepins et al., 2015). For the majority of PB reasearch, the measurement of the weight of a beehive is carried out in the following way: four load cells, which rely on mechanics and resistive theory, placed at each corner of the hive, integrated with amplifier modules that are linked to the microcontroller (Anuar et al., 2019, for example)….

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A method for mining combined data from in-hive sensors, weather and apiary inspections to forecast the health status of honey bee colonies

2020, Computers and Electronics in Agriculture

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Effect of the climate change on honey bee colonies in a temperate Mediterranean zone assessed through remote hive weight monitoring system in conjunction with exhaustive colonies assessment

2019, Science of the Total Environment

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B+WSN: Smart beehive with preliminary decision tree analysis for agriculture and honey bee health monitoring

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Honey bee colonies remote monitoring system ↗

2017, Sensors (Switzerland)

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